

Saber Ayatollahi, CFA, MSc

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SUMMARY

Quantitative risk modelling professional – applying for the Manager / Senior Manager, Quantitative Market Risk Models role – with ~7 years across model development, model validation, and sign-off for multi-asset institutional clients. At Moody's Analytics, holds delegated sign-off authority on derivatives valuation, sensitivity, and risk outputs for portfolios of \$5-25bn per engagement. Hands-on with rates, FX and inflation derivatives pricing, Monte Carlo and stochastic scenario generation, VaR/CVaR analytics, yield-curve calibration, and Python. CFA charterholder; dual MSc in Financial Modelling and Chemical Engineering (Western University), providing genuine PDE and numerical-methods fluency.

CORE SKILLS

Derivatives Pricing – Rates, FX & Inflation • VaR / CVaR Analytics & Stress Testing • Monte Carlo & Stochastic Simulation • Yield Curve Construction & Calibration • Model Development & Model Validation • PDE & Numerical Methods (Financial Engineering foundation) • Scenario Analysis & Reverse-Stress Testing • Python (Advanced), SQL, MATLAB • Model Governance & Independent Review

EXPERIENCE

Moody's Analytics

Assistant Director, Modelling Services | Toronto, May 2023 – Present

Derivatives Valuation, Calibration & Sensitivity Validation

- Holds delegated sign-off authority on valuation, sensitivity, and risk outputs delivered to multi-asset institutional client portfolios ranging \$5-25bn per engagement, cumulatively covering ~\$50bn across the book of work.
- Validates derivatives pricing outputs across interest rate, FX, and inflation products – reviewing calibration, cross-asset interactions, and sensitivity consistency at portfolio-level aggregates prior to production release.
- Reviews yield curve construction and spread calibration; escalates outputs lacking mathematical or economic defensibility, including handling edge cases such as short-end curve inversion.

Market Risk Analytics, Stress Testing & Scenario Analysis

- Reviews interest rate risk and duration analysis under parallel and non-parallel rate shocks, applying market-risk measurement frameworks analogous to Basel IRRBB standards.
- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios – providing forward-looking risk visibility across multi-period horizons.
- Architected configurable time-bucketed analytics (T+1 through multi-year), embedding behavioral cash flow assumptions, prepayment logic, and macro stress overlays.

Model Governance, Python Development & Stakeholder Communication

- Operates within Moody's formal model-governance framework – independent model validation, review and challenge, assumption validation, documentation, and escalation aligned with SR 11-7 model-risk standards.
- Re-engineered manual spreadsheet workflows into scalable, auditable Python analytics pipelines with embedded logging and validation controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) for code generation, validation scaffolding, and anomaly detection – reduced development cycle time an estimated 30-40% on comparable modules.
- Prepares analytical summaries for Heads of Risk and senior stakeholders on rates exposure, scenario impacts, and balance-sheet sensitivities; mentors junior colleagues on validation methodology.

Client Service Specialist | Toronto, May 2022 – May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform; advised on asset, liability, and risk model configurations and validated outputs post-deployment.
- Translated client investment-team requirements into structured Product Owner specifications; contributed to full migration of all assigned accounts from legacy Calypso environment to PFaroe PM platform.

Ernst & Young (EY)

Senior Consultant, IFRS & Financial Risk Transformation | Toronto, Sep 2021 – Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurance clients; advised actuarial, IT, finance, and program-management functions on standard adoption, governance documentation, and regulatory-readiness processes.
- Developed IFRS 17 advisory product offerings and go-to-market collateral; coordinated cross-functional milestone delivery across actuarial, risk, finance, IT, and PM stakeholders.

Ortec Finance Canada

Consultant, Asset-Liability Management & Risk | Toronto, Feb 2019 – Sep 2021

- Built and interpreted stochastic economic scenario generators using Monte Carlo simulation to assess interest rate, inflation, and currency risk under base-case and stressed environments for institutional clients.
- Performed asset-only and asset-liability (surplus) portfolio optimization based on VaR and CVaR via Ortec's GLASS platform; conducted risk decomposition, contribution-to-risk attribution, and near-optimal frontier robustness analysis.
- Delivered ALM and liability-driven investment (LDI) studies for Canadian and international pension funds; modelled actuarial liabilities alongside all major investment-product types and economic variables.
- Performed cash flow, liquidity, and hedging analysis across interest rate, inflation, and FX exposures for international clients; presented findings to investment committees at pension-fund clients on-site.
- Led onboarding and the ALM study for the University Pension Plan (UPP) merger of three single-employer plans into a new jointly-sponsored pension plan – team of three developing and validating the model, evaluating duration, inflation, currency, and leverage overlay strategies.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016